

Answer: D

- 38** The main charge carriers in n-type and p-type semiconductor are electrons and holes respectively. In the formation of a depletion region, the electrons from the n-type diffuse towards the p-type conductor. During forward bias, current flows from the p-type to the n-type semiconductor, which means the holes diffuse from the p-type to the n-type semiconductor.

Answer: C

- 39** $N = N_0 e^{-\lambda t}$